

RF Link Planner

Terrain-aware network planning for Mesh Rider radios. What the tool does, how far we can trust it, and what it has already found.

doodlesim.jus2419497.workers.dev Live now - no login, nothing to install

\$0 / mo Hosted on our own Cloudflare account	14 radios Full per-MCS data, including v4 and RM-1300/1400	90 antennas Every row audited against manufacturer datasheets	Exact Reproduces the official Doodle Labs estimator
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In one paragraph

We have an in-house planning tool that puts Mesh Rider links on a real map. You place radios, pick the actual radio, band, antenna and mast height, and it tells you what the link will really do - throughput, margin, coverage footprint, and where it breaks. Underneath it reproduces the official Doodle Labs range estimator exactly, then adds the terrain that a distance-only calculator cannot see. It is internal-only today. Two independent checks sit behind it: a field result reported from Ukraine, and a separate, unrelated batch of customer radio logs.

What you can do with it

- **Plan a link.** Place radios and get distance, best usable MCS, throughput and margin. Click a link for the terrain cross-section with the Fresnel zone drawn, so you can see which ridge is costing you.
- **Coverage maps, two ways.** **Terrain-following** for UGVs, vehicles and handhelds - what can something driving around here reach? And **flight level** for UAS, where you drag an altitude slider and coverage repaints live. It also maps the minimum altitude a drone needs to connect, point by point.
- **Mesh overlap and redundancy.** Combines every radio footprint and colours by how many radios cover each spot - one, two (survives any single outage), or three or more. Reports the area of each in square kilometres.
- **Mission routes.** Paste a waypoint list or click it on the map, set up the vehicle radio, and get percentage of route with a link, percentage at your required data rate, longest dropout in metres, handover count, and which radio serves which stretch.
- **Plan Advisor.** Describe the mission - "ground station to UAV, 10 km, 5 Mbps" - and it searches every radio, band, channel width and antenna combination, then hands back ranked builds with aiming, heights and cabling. It will also extend an existing network and site a relay when one hop cannot reach.
- **Locate by signal.** Nano and Mini radios carry no GPS. Use the OEM and Wearable units, which do, as anchors: enter the levels they report and the tool works out position from the link budget, with an honest confidence circle rather than a false pin.
- **Customer-ready reports.** One click exports a document with the map, equipment table, per-link terrain analysis and minimum UAV altitudes. Prints straight to PDF.
- **Fade margin you control.** Set 10 dB for our standard plan, 12-14 dB for field practice, higher for contested RF. Every calculation follows it and the report states the value used, so nothing goes out based on a looser number

than you quoted.

Why the numbers can be trusted

The link budget **reproduces the official Doodle Labs Range Estimation Tool exactly** - same conservative sensitivities, same per-MCS transmit powers - verified to the metre across roughly thirty test cases. On top of that it adds what a distance-only calculator cannot: real elevation data, earth curvature, Fresnel-zone clearance, diffraction, antenna radiation patterns and frequency-dependent cable loss.

The antenna audit

Our recommended-antenna spreadsheet was checked row by row against manufacturer datasheets. That work is now baked into the tool:

Result	Count	What it means
Confirmed	31	Sheet matched the manufacturer datasheet
Specs corrected	19	Real product, wrong numbers on our sheet
Partly verified	14	Some fields confirmed; the rest unpublished or self-contradictory
Not a real product	20	Family labels, radio platforms, or part numbers that do not exist
Unverifiable	6	No authoritative source found anywhere
Total	90	Every row on the sheet

Every one of the 90 is listed individually in the appendix.

Four catches worth knowing about

Most of the L-com HG4958-xx part numbers do not exist as written. Both Poynting entries are cellular antennas we had listed as Wi-Fi. The Triad BDA is rated 25 W, not the 16 W on our sheet. And one recommended sector antenna, rated 5 W maximum input, was paired with a 20 W amplifier that would have destroyed it.

Tested against reality

Check 1: Ukraine field test (colleague-reported figures only)

A colleague working in country gave us the values he had on hand: a 24 dBi parabolic on the ground station at 2 m, a 3 dBi multipolarised omni on the aircraft at 1000 m, a 5 MHz channel swept from 1675 to 2500 MHz, clear air with no jamming, and the link dropping out at 42-44 km. He works to a 12-14 dB fade margin. Fed only those numbers, the tool puts the cliff inside his exact window with no adjustment:

Fade margin	Tool prediction at 42.6 km
10 dB	MCS 10, 7.4 Mbps, +10.5 dB margin
12 dB	MCS 9, 5.1 Mbps, +12.5 dB margin
14 dB	MCS 1, 2.7 Mbps, +15.5 dB margin

Fade margin	Tool prediction at 42.6 km
20 dB	No link

What this check is, and is not

These are the only figures we have from that site - his stated kit, heights, channel and the distance at which he lost the link. There are no logs and no measured signal levels from Ukraine, so this is a consistency check against what he reported, not an independent measurement. It is still meaningful: the model was not tuned to match, and it landed in his window first time.

It also gave him two things to act on. His frequency sweep up to 2500 MHz costs about 3.5 dB, so staying near 1675 MHz is worth roughly 1.5 times the range. And the tightest point on the whole 43 km path is 200 m in front of his ground station, where a 2 m antenna only just clears - lifting that dish is the cheapest margin he can buy.

Check 2: what it found in customer logs (a separate, unrelated batch)

We built parsers for Mesh Rider support bundles and ran six of them, from three unrelated customer systems, about 28,000 telemetry samples. These are an arbitrary sample of logs and have **nothing to do with the Ukraine test above** - they are a second, independent check that the tooling finds real faults:

System	Finding
Flight 7 ground station	One receive chain 10 dB down (chains at -73 and -63 dBm). The link stayed associated the whole time yet was unusable: 80% packet loss, never got past MCS 6, because a dead chain means the second data stream never works. Signal strength was never the problem.
Flight 7 channel	The 918 MHz channel was busy 70% of the time with noise peaks to -75 dBm, against 9% on a healthy reference pair.
Kratos fleet	The same 5-7 dB chain imbalance on both the ground station and the relay, against every peer - which localises it to their own antenna paths rather than the air.
Kratos relay	Linked only 69.6% of the time while hopping across three frequencies and two channel widths, against a ground station that never left one channel.

It also caught its own false alarms

A "transmit power is only 15 dBm" that turned out to be the 5 GHz Wi-Fi hotspot interface, not the mesh radio (which runs 30-32 dBm as configured). A "98.9% error rate" that was simply spectral-scan samples being counted as errors. And one of our own conclusions withdrawn after a timing audit showed two logs were captured 32 minutes apart rather than simultaneously. Being able to rule things out matters as much as finding them.

Where it is honest about its limits

- Terrain data is about 30 m resolution, so **buildings and trees are not modelled**. Urban and forest predictions run optimistic - budget extra margin there.
- Interference and rain are covered only by the fade margin, not modelled explicitly.
- Predictions are planning-grade engineering estimates, not a warranty. Field-verify critical links.

- The core link budget is thoroughly validated. The newer features - coverage modes, routes, signal-based ranging - are self-audited but have **not had independent review** yet.
- Radios without GPS can only be positioned when several of them measure each other **at the same time**. Sequential captures cannot be combined, and the tool refuses rather than guessing.

Where to find more

Resource	Link
The tool itself	doodlesim.jus2419497.workers.dev
Source, data and all documentation	github.com/jus24194971/DoodleSim
Antenna verification, all 90 rows	data/verification_report.md
Customer log findings, six bundles	data/LOG_FINDINGS.md
Ukraine field validation	data/FIELD_VALIDATION_UA.md
Technical summary, for engineering review	DoodleSim-Technical-Summary.pdf

Repository links need access to the DoodleSim repo; the tool itself needs nothing.

What would help most

Have a play - a 60-second tour runs on your first visit, and Help has tutorials and an FAQ. Then tell us where it is wrong, especially anywhere it disagrees with something you have actually measured: hit **Save** and send the .json with what you saw. Access control goes in before anything reaches a customer.

Appendix: every antenna checked

All 90 rows of the recommended-antenna sheet, each audited against the manufacturer's own datasheet. Where our sheet and the datasheet disagree, both figures are shown. Source links for every row are in [data/verification_report.md](#).

Confirmed 31 of 90

Our sheet matched the manufacturer datasheet on band, gain, pattern and type.

#	Manufacturer and model	Band (verified)	Gain: sheet / verified	Note
1	Alfa Network APA-M04	2.4-2.5 GHz	7	
2	Antenna Experts AY-2450	2300-2600 MHz	13	
3	Antenna Experts JC10-2450	2400-2500 MHz	10	
4	Antenna Experts JC12-2450	2400-2500 MHz	12	
5	Antenna Experts (India) AH-4959	4900-5900 MHz	16 dBic	
6	Cisco AIR-ANT2422DW-R	2.4-2.484 GHz	2.2	

#	Manufacturer and model	Band (verified)	Gain: sheet / verified	Note
7	D-Link ANT24-0700	2.4-2.5 GHz	7	
8	Hascall-Denke FXDP4.4-6.0-6-D	4400-6000 MHz	6	
9	L-com (HyperLink) HG2403CU	2400-2500 MHz	3	
10	L-com (HyperLink) HG2424EG	2400-2500 MHz	24	
11	MARS Antennas MA-WO56-DP10	4.9-5.9 GHz	10	
12	MikroTik mANT30	4.7-5.875 GHz	30	
13	MikroTik mANT30PA	4.7-5.875 GHz	30	
14	Pasternack Enterprises PE51078	2400-2500 MHz	5.5	
15	Select Antenna (Select Telecom) SEL-SEC-5816M	4900-6000 MHz	2×16	
16	Smart Electronics Communication (SEC Antenna) SGP-4/5/6/8 series	700-3000 MHz (all four models)	16-34	
17	Telimart India Pvt. Ltd. TM55D-HVOMNI-12	4900-5900 MHz	12	
18	Telimart India Pvt. Ltd. TM55D-HVSCTR-16	4.9-5.9 GHz	16	
19	TP-Link TL-ANT2409A	2.4-2.5 GHz	9	
20	TP-Link TL-ANT2424B	2.4-2.5 GHz	24	
21	Ubiquiti AM-2G15-120	2.3-2.7 GHz	15	
22	Ubiquiti AM-5G19-120	5.15-5.85 GHz	19	
23	Ubiquiti AM-5G20-90	5.15-5.85 GHz	20	
24	Ubiquiti AMO-2G10	2.35-2.55 GHz	10	
25	Ubiquiti AMO-5G10	5.45-5.85 GHz	10	
26	Ubiquiti RD-5G30-LW	5.1-5.9 GHz	30	
27	Ubiquiti RD-5G34	5.1-5.9 GHz	34	
28	Verdant Telemetry & Antenna Systems Pvt. Ltd. JD 201	110 MHz - 400 MHz	-	
29	Verdant Telemetry & Antenna Systems Pvt. Ltd. JD 322 N	30 MHz - 550 MHz	-	

#	Manufacturer and model	Band (verified)	Gain: sheet / verified	Note
30	Verdant Telemetry & Antenna Systems Pvt. Ltd. JD 401	20 MHz - 1200 MHz	-	
31	Verdant Telemetry & Antenna Systems Pvt. Ltd. JL 50	1430 MHz - 1540 MHz	-	

Specs corrected 19 of 90

A real product, but our sheet had it wrong. The tool carries the verified figures.

#	Manufacturer and model	Band (verified)	Gain: sheet / verified	Note
32	Alfa Network AOA-2458-59-TF	2.4-2.5 GHz and 5.1-5.9 GHz	9 / 5 dBi @ 2.4 GHz; 9 dBi @...	Model suffix 59 = 5/9 dBi: 9 dBi applies only to 5 GHz; 2.4 GHz is 5 dBi. Sheet's flat 9 overstates 2.4 GHz...
33	Aruba (HPE) AP-ANT-13B (JW001A)	2.4-2.5 GHz and 4.9-5.9 GHz	4 / 4.4 dBi @ 2.4 GHz; 3.3 dBi...	Gain is per-band (4.4/3.3 dBi), not a flat 4 dBi. Legacy 802.11a/b/g/n accessory (AP-124/AP-204 era); aruba...
34	Celestial (Celestial Space Technologies) CST-S-9882 (sheet: 'Celestial S-Band Patch')	2.05-2.30 GHz (190 MHz BW)	6-8 / 6-8 dBi	Band wider than sheet claims: 2.05-2.30 GHz per satsearch. Gain/pol/type confirmed.
35	Cisco AIR-ANT2524V4C-R	2.4 GHz and 5 GHz dual-band	4 / 2 dBi @ 2.4 GHz; 4 dBi @...	Sheet's single 4 dBi figure is 5 GHz only; 2.4 GHz is 2 dBi. Cisco.com blocked fetch (403); specs cross-che...
36	Gleam Light India Pvt. Ltd. COMNIID040 / WIFIFOMID2	COMNIID040: 700-2700 MHz; WIF...	2 / COMNIID040: 4-5 dBi; WIF...	Row merges two different Gleam Light products; both exist on manufacturer pages, but sheet gain (2 dBi) mat...
37	Gleam Light India Pvt. Ltd. GLI/OMNI/01	700-2700 MHz	5-7 / 2 dBi	Manufacturer page lists GLI/OMNI/01 as a 2 dBi indoor omni, 700-2700 MHz. Sheet gain 5-7 dBi is wrong; band...
38	Hascall-Denke FXPR1350-2700-D	1350-2700 MHz	26 / 20-26 dBi +/-0.5 across b...	Sheet lists only top-of-range gain; datasheet says 20-26 dBi. Isolation 32-45 dB; parabolic dish (P/N 1Y282...
39	INPAQ Technology ACA-8010-A1-MC-S	2400-2500 MHz (center 2450)	2.5 / 1.9 dBi peak	Not an Eteily product: 'ET-' prefixed model not found anywhere. Real part is INPAQ ACA-8010-A1-MC-S. Gain 1...
40	MikroTik mANT15s	5.17-5.825 GHz (5 GHz only)	15 / 15 dBi	Sheet band 2.4/5 GHz is wrong; MTAS-5G-15D120 is 5 GHz only. Sheet URL slug (mant15s) now 404s; current pag...
41	MikroTik mANT19s	5.17-5.825 GHz (5 GHz only)	19 / 19 dBi	mANT19s denotes the standalone antenna (MTAS-5G-19D120), not the integrated mANTBox 19s radio (RB921GS-5HPa...
42	Poynting OMNI-402	410-3800 MHz (wideband cellul...	3 / 6.2 dBi peak (varies acro...	EOL/discontinued per Poynting (successors OMNI-493/OMNI-414). It is an LTE/5G marine omni, not a 2.4/5 GHz...
43	Poynting OMNI-600	410-3800 MHz; does NOT cover...	7 / 6.2 dBi peak	EOL/discontinued per Poynting (alternatives OMNI-214/OMNI-293/XPOL-1-5G). Cellular antenna topping out at 3...
44	Poynting XPOL-2-5G	617-4200 MHz (cellular 2G-5G;...	3.5 / up to 11 dBi (band-depend...	Cellular LTE/5G antenna, not WiFi. Same model appears in row 69 with a different claimed band; datasheet ra...

#	Manufacturer and model	Band (verified)	Gain: sheet / verified	Note
45	Poynting XPOL-2-5G	617-4200 MHz; upper limit 420...	4 / up to 11 dBi (band-depend...	Duplicate of row 35 with conflicting claims. Datasheet: cellular antenna, 617-4200 MHz max; unusable for RM...
46	SEC Antenna (Smart Electronics Communication) SSP58SPG29	5150-5850 MHz	29 / 29 dBi	Two corrections: manufacturer is SEC Antenna (secantenna.com, Smart Electronics Communication), NOT Select...
47	Select Antenna (Select Telecom) SEL 24-17	2400-2500 MHz	17 / 17 dBi	Model written as SEL-2417 in sheet; official page lists it as 'SEL 24-17'. Page's 5/7 deg beamwidths look i...
48	Southwest Antennas 1001-126	4.4-6.0 GHz	1.5 / 1.5 dBi (official page ti...	Sheet pattern wrong: 1001-126 is an omni spring-base dipole, not a directional panel. Gain conflict noted (...)
49	Southwest Antennas 1009-015	1.625-2.5 GHz	12 / 12 dBi max	Sheet pattern wrong: this is a directional panel (55°/32° HPBW), not an omni whip. Band/gain/vertical pol c...
50	Verdant Telemetry & Antenna Systems Pvt. Ltd. JD 350 B	30-400 MHz plus 960-1240 MHz...	- / Not stated on product page	Official Verdant page used instead of ec21 reseller link. Sheet omits the 960-1240 MHz L-band; 100W CW, 4kW...

Partly verified 14 of 90

Some fields confirmed; the rest is unpublished, marketplace-grade, or the vendor contradicts itself. Treat the unconfirmed fields with care.

#	Manufacturer and model	Band (verified)	Gain: sheet / verified	Note
51	Antenna Experts JCM6-2450	2400-2500 MHz	6 / 6 dBi (title) / 9 dBi typ...	Datasheet contradicts itself: header says 6 dBi, electrical spec table says 'Gain - Typical 9 dBi'. Sheet's...
52	Celestial (Celestial Space Technologies) CST-L-7070	L-band (exact MHz range not p...	-	Model exists; satellite TT&C patch, full-duplex CCSDS. Gain/pol/freq details only in gated datasheet. S...
53	CSG Networks CSG 62411 ODN	2.4 GHz (per official datashe...	11	csgnets.com unreachable (connection drops on http and https; also via browser). Model exists - datasheet fi...
54	CSG Networks (REN brand); design equiv. Madhu Subtronic MES REN 65817 BFN	5725-5850 MHz (equivalent MES...	- / 17 dBi	csgnets.com (REN seller) unreachable and archive blocked; specs taken from identically-numbered MES 65817 B...
55	Eteily Technologies ET-5812FG5-NMS	5.8 GHz nominal (exact range...	12 / 12 dBi	Official page confirms model, 12 dBi, N-Male, fiberglass. Marketed as 5.8 GHz; sheet's 5100-5800 range not...
56	Eteily Technologies ET-91518Y2-2L20C-NFS	915 MHz nominal (exact range...	18 / 18 dBi	Official page confirms model, 18 dBi, yagi, N-F connector. Sheet band 824-960 not stated; page markets it a...
57	Eteily Technologies ET-9153R-SMMO	915 MHz nominal (page states...	3 / 3 dBi	Official page confirms model, gain, type, connector; marketed as 915 MHz. Sheet's 824-960 range not stated...
58	Gleam Light India Pvt. Ltd. OMNIOD4120	824-960 MHz (OMNIODA120); pag...	10-12 / 12 dBi	OMNIOD4120 as written not found; almost certainly a typo of OMNIODA120 (824-960 MHz, 12 dBi). The 1710-2700...
59	Gleam Light India Pvt. Ltd. OMNIODA070	860-870 MHz LoRa (listed 800-...	7 / 7 dBi	Exact code OMNIODA070 not found anywhere; Gleam Light's OMNIODA coding is real (OMNIODA120 exists) and a ma...

#	Manufacturer and model	Band (verified)	Gain: sheet / verified	Note
60	Select Antenna (Select Telecom) SEL 24-24	2400-2500 MHz	24-30 (spec... / 24 dBi (product title) vs...	Sheet's noted 24-30 conflict is real: official page titled 24 dBi but spec table says 30 dBi. 24 dBi is phy...
61	Southwest Antennas 1004-005	4.0-6.4 GHz (C-band)	9 / 9 dBi	Removed from southwestantennas.com (404). RFMW: 'Antenna, Panel, 4.0-6.4 GHz, 9dBi, SMA(F), Kydex, Side Mou...
62	Telimart India Pvt. Ltd. TM26D-HVPANEL-14	2.3-2.9 GHz	14 / 8 dBi (TradeIndia listing...	Model exists on Telimart's own TradeIndia store. Gain conflicts: suffix implies 14 dBi but listings state 8...
63	Telimart India Pvt. Ltd. TM26L-SPDISH-15/21	2.3-3.9 GHz (TM26L SPDISH fam...	15-21 / Variants SPDISH-21/-27/-3...	'-15/21' is a gain-range family label, not a SKU. TM26L SPDISH family confirmed via Telimart TradeIndia lis...
64	Telimart India Pvt. Ltd. TM55L-SPDISH-23/38	4.9-5.9 GHz (dish family)	23-38 / Family SKUs found at 26,...	'-23/38' is a gain-range family label, not a SKU. Telimart 4.9-5.9 GHz dish family (up to 38 dBi/6 ft) conf...

Not a real product 20 of 90

Family labels, radio platforms rather than antennas, or part numbers that do not exist as written. These should come off the sheet.

#	Manufacturer and model	Band (verified)	Gain: sheet / verified	Note
65	Astra Microwave Products AMP-PATTS-203 (Example)	S-Band (Telemetry ~2.4 GHz)	-	Model is explicitly labeled '(Example)' and no 'AMP-PATTS-203' exists in any search result. Astra MWP does...
66	Astra Microwave Products Ltd. (Portfolio)	1-18 GHz (various)	Varies	Portfolio row, no specific model. Astra Microwave (Hyderabad, est. 1991) is a real defense/space RF house;...
67	Cambium Networks 450i	5.0 GHz	17	PMP 450i is a radio platform, not an antenna. Sheet URL points to one accessory (5 GHz 60° sector); the 17...
68	Cambium Networks ePMP	5.0 GHz	15	ePMP is a radio platform family, not an antenna model. The 15 dBi likely refers to the ePMP 5 GHz sector an...
69	Cobham Antenna Systems C-Band	4.4-6.0 GHz	Various	Family label only; no specific model. Cobham antenna business now trades as European Antennas/Cobham Aerosp...
70	CommScope 5 GHz	5.0 GHz	18	'5 GHz' is a frequency/family label, not a CommScope model. No product to verify; the 18 dBi figure cannot...
71	Hascall-Denke 2.2-2.5/4.4-5.875 GHz (custom models)	Custom models	Various	Family/custom-capability row; model field is a band string, not a part number. Hascall-Denke does offer S/C...
72	L-com (HyperLink) HG4958-12	4.9-5.8 GHz	12	Model as written not sold. Very likely means HGV-4958-12U: 4.9-5.8 GHz 12 dBi omni, N-Female, vertical pol...
73	L-com (HyperLink) HG4958-14	4.9-5.8 GHz	14	No 'HG4958-14' found anywhere. Real 4.9-5.8 panels are 7 (PCR), 17 (DP-090 sector), 19 (19DP), 23 (23P) dBi...
74	L-com (HyperLink) HG4958-15U	4.9-5.8 GHz	15	No 'HG4958-15U' found; L-com's 4.9-5.8 omni line (HGV-4958-xxU) tops out at 12 dBi. No 15 dBi broadband 4.9...

#	Manufacturer and model	Band (verified)	Gain: sheet / verified	Note
75	L-com (HyperLink) HG4958-18	4.9-5.8 GHz	18	No 'HG4958-18' exists anywhere (L-com, distributors, archives). Real 4.9-5.8 panels: HG4958-19DP (19 dBi du...
76	L-com (HyperLink) HG4958-19	4.9-5.8 GHz	19	Plain 'HG4958-19' not a product. Almost certainly means HG4958-19DP: 4.9-5.8 GHz 19 dBi DUAL-polarized flat...
77	L-com (HyperLink) HG4958-22D	4.9-5.8 GHz	22	No 'HG4958-22D' found. Closest real model: HG4958-22EG (22 dBi die-cast GRID, not solid dish, V/H selectabl...
78	L-com (HyperLink) HG4958-24D	4.9-5.8 GHz	24	No 'HG4958-24D' found in any source. Real 4.9-5.8 dishes: HG4958DP-25D (25 dBi), HG4958DP-30D, HG4958DP-34D...
79	L-com (HyperLink) HG4958-29D	4.9-5.8 GHz	29	No 'HG4958-29D' found. Closest: HG4958DP-30D (30 dBi dual-pol dish, 4.9-5.8) or HG5158DP-29D (28.5 dBi, 5.1...
80	Netgear ANT2408	2.4 GHz	8	No Netgear ANT2408 exists: sheet URL 404s and no vendor/reseller listing found. Sheet's own doubt confirmed...
81	Synergy Telecom Pvt Ltd 2.4 GHz Omni (12 dBi)	2400-2500	12	Generic family label, no model number. Synergy Telecom (synergyantenna.com) exists but its omni catalog sho...
82	Telemart India (reseller) (reseller)	Various	Various	Reseller row with no model number (cell truncated '(reseller)'. telemartindia.com returned HTTP 403; Telema...
83	Verdant Telemetry & Antenna Systems multi-band	1.6-2.5 GHz & 4.4-6.0 GHz	Various	Family/portfolio row only; no specific model number to verify.
84	- 2.4/5.8 GHz Whip	2400/5800	3	Generic dual-band whip category with no manufacturer, model number, or datasheet URL; not a specific verifi...

Unverifiable 6 of 90

No authoritative source could be found. Not necessarily wrong, but nothing supports it.

#	Manufacturer and model	Band (verified)	Gain: sheet / verified	Note
85	Benelec 2759 (02759)	4.9-5.825 GHz	10	Sheet URL dead (benelec.com.au cert now benelec.au; /product/02759/ 404s). No 02759 in Benelec's current ca...
86	CSG Networks (REN brand) REN 65824 SGPN	5725-5825	-	Model appears only as a dead csgnets.com URL; site unreachable, no spec sheet found. Numbering convention (...)
87	L-com (HyperLink) HG2405U	2.4 GHz	5	Plain 'HG2405U' not sold; real variants are HG2405UP-NF (5 dBi outdoor omni, N-F) and HG2405U-NMO / HG2405U...
88	L-com (HyperLink) HG2406CU	2.4 GHz	6	No 'HG2406CU' found on L-com or any distributor; ceiling CU line only has HG2403CU (3 dBi). HGV-2406U is a...
89	L-com (HyperLink) HG2410U	2.4 GHz	10	'HG2410U' as written not in current L-com catalog (likely discontinued legacy HyperLink number). Closest cu...

#	Manufacturer and model	Band (verified)	Gain: sheet / verified	Note
90	L-com (HyperLink/HyperGain) HG2415P	2.4 GHz	15	No L-com product 'HG2415P' found; sheet URL 404s. Real models: HG2415P-180 (15 dBi 180-deg sector, vertical...